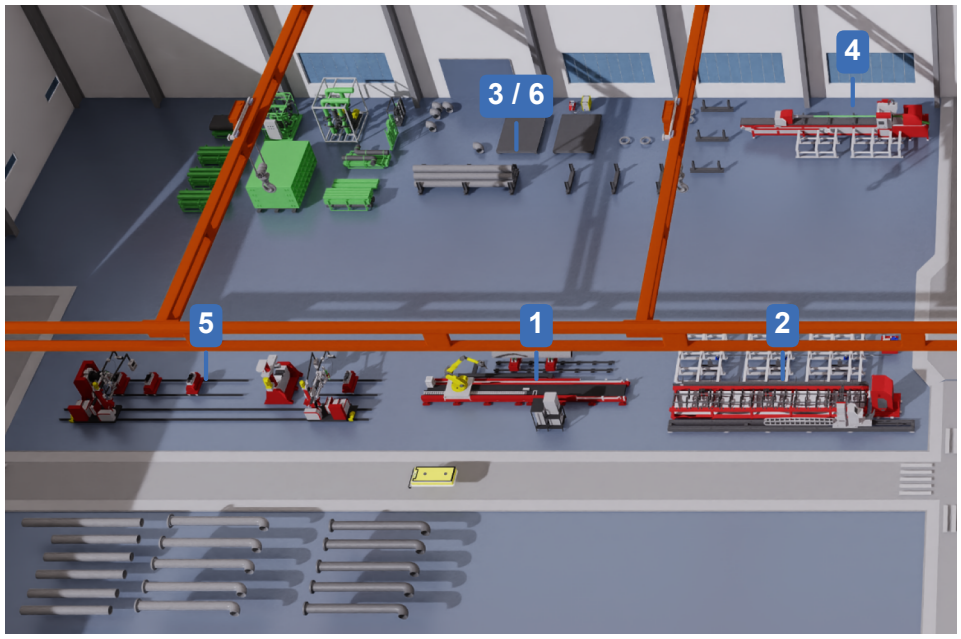
(a) Digital-twin production environment



1	CNC pipe cutting
2	Pipe grooving
3 / 6	Shared workbenches
4	Robotic root welding
5	Rotary surface welding

Overhead gantries • transport robots
Workers • storage / buffers

Left-side detail of the full scene

(b) Six-step manufacturing route and handling dependencies

Raw pipe

Pipe + flange + elbow

Finished water pipe

01	02	03	04	05	06
Pipe cutting	**Pipe grooving**	**Assembly & spot welding**	**Argon-arc root welding**	**MIG surface welding**	**Rust-proof painting**
Cutting machine	Grooving machine	Workbench	Welding robot	Rotary welder	Workbench

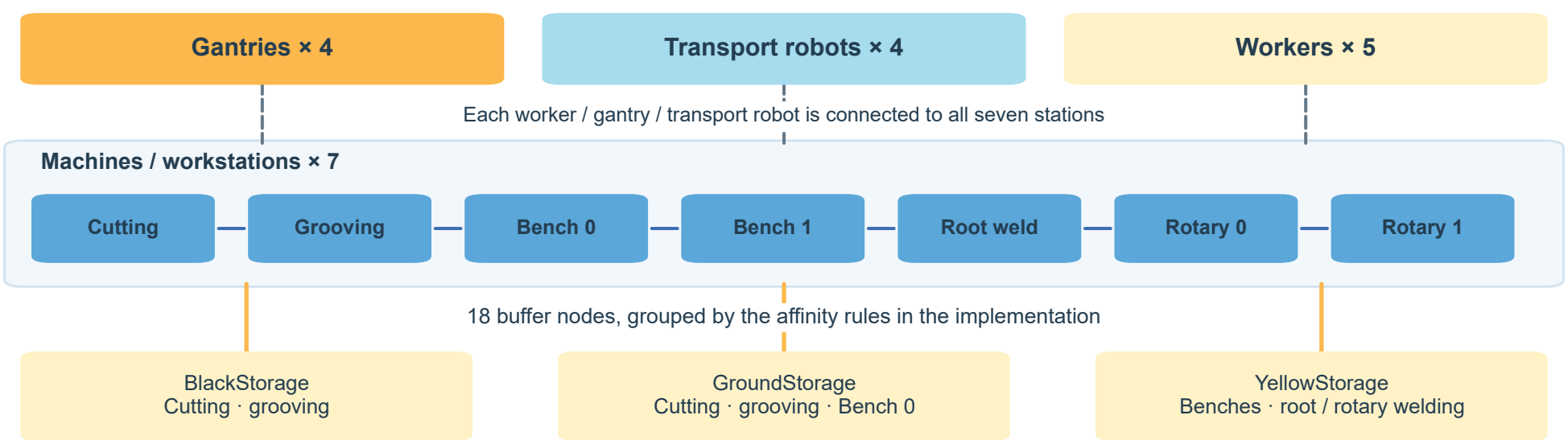
Steps 3 and 6 reuse the workbench resource family.

Material transfer between processing steps

Same zone: worker-assisted gantry transfer

Cross zone: source gantry → transport robot → destination gantry

(c) Multi-entity structural dependencies used by the model



Solid: station / buffer relations Dashed: bundled agent–station relations

Undirected, grouped view; within-type gantry / robot links and self-loops omitted.